Tianze Xu

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Andreasstrasse 15, 8050 Zurich, Switzerland

Education

Aug 2025–present **PhD in Computational Linguistics and Phonetics**
University of Zurich (UZH), Switzerland
- Supervisors: Professor Dr Volker Dellwo, Dr Elisa Pellegrino, & Professor Pascal Belin (Aix-Marseille University, France)
- Project: *Attention to vocal identity cues* (Part of the Marie Skłodowska-Curie Actions Doctoral Network *Voice Communication Sciences*)

Oct 2024–Jun 2025 **MPhil in Theoretical and Applied Linguistics**
University of Cambridge, United Kingdom
- Supervisor: Dr Kirsty McDougall
- Grade: Distinction (84%)
- Dissertation: *Perceived voice similarity within and across languages: A study of Cantonese–English bilingual speakers*

Sep 2020–Jun 2024 **BA in Language Science &**
Honours Programme in International Organisation Studies
Shanghai International Studies University (SISU), China
- Supervisor: Professor Xiaoming Jiang
- Cumulative GPA: 3.95/4
- Thesis: *Is thematic role assignment syntactic or semantic in nature? Evidence from computational modelling on naturalistic fMRI*

Funding & Awards

Jun 2026–May 2029 **Digitalization Initiative of the Zurich Higher Education Institutions Innovation Program Project Call Grant**
University of Zurich & Zurich University of Applied Sciences
- Project: *Fake or real? Exploring brain signals for deepfake voice detection* (#50)
- Role: Contributor
- Amount: CHF 255,000 (£241,640)

Aug 2025–Jul 2028 **Marie Skłodowska-Curie Actions Doctoral Networks Grant, Horizon Europe**
European Research Executive Agency & Swiss State Secretariat for Education, Research and Innovation
- Project: *Voice Communication Sciences* (#101168998)
- Role: Fellow
- Amount: €303,567 (£263,170)

Oct 2025–Jun 2026 **Research Travel Grants**
Faculty of Arts and Social Sciences, University of Zurich
- Role: Lead researcher
- Amount: CHF 3,000 (£2,830)

Dec 2024–Jun 2025 **Postgraduate Travel and Research Grant**
Christ's College, University of Cambridge

- Role: Lead researcher
- Amount: £1,650

Jun 2024 **First Class Honours Scholarship**
Shanghai International Studies University
- Amount: CN¥ 50,000 (£5,550)

Sep 2020–Apr 2024 **National Undergraduate Training Programme for Innovation and Entrepreneurship Grant**
Shanghai International Studies University
- Project: *Predicting voice discrimination ability from EEG signals* (#202310271021)
- Role: Lead researcher
- Amount: CN¥ 18,400 (£2,050)

Mar 2021–Mar 2024 **Outstanding Student Awards & Scholarships**
Shanghai International Studies University
- Amount: CN¥ 8,800 (£980)

Oct 2023 **Honours College Overseas Exchange Scholarship**
Shanghai International Studies University
- Amount: CN¥ 10,000 (£1,120)

Aug 2023 **Korea University Global e-School Programme Scholarship**
Korea Foundation, Ministry of Foreign Affairs of South Korea
- Amount: US$ 2,850 (£2,150)

Publications

Preprint **Xu, T.**, Jiang, X., & Dellwo, V. Acoustic cue weighting in voice discrimination varies across words andsentences and by speaker gender. PsyArXiv. https://doi.org/10.31234/osf.io/4zrwy_v4

Accepted Chen, Y., **Xu, T.**, Farhadipour, A., Burdisso, S., Cyrta, P., Dellwo, V., Madikeri, S., Marxer, R., & Motlicek, P. EmoPilot: Label-free emotion cue prediction for zero-shot text-to-speech. *2026 IEEE Spoken Language Technology Workshop (SLT)*.

Accepted Dai, L., Huang, Y., **Xu, T.**, Luo, R., & Jiang, X. Resting-state alpha power and cross-hemispheric beta connectivity are associated with individual differences in voice identity perception. *Journal of Speech, Language, and Hearing Research*.

2026 **Xu, T.**, & Jiang, X. Voice perception across languages: Reduced similarity between speakers, stable identity within speakers. *Journal of Speech, Language, and Hearing Research*, *69*(11), 5401–5427. https://doi.org/10.1044/2026_JSLHR-25-01029

2026 **Xu, T.**, & Dellwo, V. Learning speaker identities in dialogue: Conversational familiarisation modulates response bias and confidence in voice recognition. In *Proceedings of Interspeech 2026* (pp. 5889–5893). https://doi.org/10.21437/Interspeech.2026-1174

2026 **Xu, T.**, Li, X., Jiang, X., & Dellwo, V. Human-like cross-language generalisation in deep neural speaker embeddings and its acoustic foundations. In *Proceedings of Interspeech 2026* (pp. 2313–2318). https://doi.org/10.21437/Interspeech.2026-858

2026 **Xu, T.**, Jiang, X., & Dellwo, V. Dominant role of temporal–prosodic cues in sentence-level voice discrimination: Insights from interpretable machine learning. In *Proceedings of Speech Prosody 2026* (pp. 609–613). https://doi.org/10.21437/SpeechProsody.2026-123

2025 **Xu, T.** A critical review of voice-space models. *Cambridge Occasional Papers in Linguistics*, *17*, 22–36. https://www.copil.mmll.cam.ac.uk/files/v17_Xu.pdf

2025 **Xu, T.**, Jiang, X., Zhang, P., & Wang, A. Introducing the Sisu Voice Matching Test (SVMT): A novel tool for assessing voice discrimination in Chinese. *Behavior Research Methods*, *57*(3), Article 86. https://doi.org/10.3758/s13428-025-02608-3

2025 **Xu, T.**, Li, J., & Jiang, X. Semantic processing of argument structure during naturalistic story listening: Evidence from computational modeling on fMRI. *NeuroImage*, *314*, Article 121253. https://doi.org/10.1016/j.neuroimage.2025.121253

Conference Presentations

Dec 13–16, 2026 Chen, Y., **Xu, T.**, Farhadipour, A., Burdisso, S., Cyrta, P., Dellwo, V., Madikeri, S., Marxer, R., & Motlicek, P. EmoPilot: Label-free emotion cue prediction for zero-shot text-to-speech [Poster presentation]. 2026 IEEE Spoken Language Technology Workshop, Palermo, Italy.

Sep 29–Oct 2, 2026 Mauchand, M., **Xu, T.**, Dellwo, V., & Hervais-Adelman, A. *Dissecting the emergence of vocal identity representations across conversations* [Poster presentation]. 18th Annual Society for the Neurobiology of Language Meeting, Geneva, GE, Switzerland.

Sep 29–Oct 2, 2026 **Xu, T.**, Yao, M., Hervais-Adelman, A., Giroud, N., Dellwo, V., & Elmer, S. *Hearing speech across voices: EEG evidence for the role of speaker variability in statistical learning of word boundaries* [Poster presentation]. 18th Annual Society for the Neurobiology of Language Meeting, Geneva, GE, Switzerland.

Sep 27–Oct 1, 2026 **Xu, T.**, & Dellwo, V. *Learning speaker identities in dialogue: Conversational familiarisation modulates response bias and confidence in voice recognition* [Oral presentation]. Interspeech 2026, Sydney, NSW, Australia.

Sep 27–Oct 1, 2026 **Xu, T.**, Li, X., Jiang, X., & Dellwo, V. *Human-like cross-language generalisation in deep neural speaker embeddings and its acoustic foundations* [Oral presentation]. Interspeech 2026, Sydney, NSW, Australia.

Sep 2–4, 2026 Mauchand, M., **Xu, T.**, Varano, E., Hervais-Adelman, A., & Dellwo, V. *Is voice identity acoustically more pronounced around dialogue turn boundaries?* [Poster presentation]. 3rd Interdisciplinary Conference on Voice Communication, Jena, TH, Germany.

Sep 2–4, 2026 **Xu, T.**, & Dellwo, V. *Conversational coherence shapes voice learning: Dialogue-based exposure improves voice recognition sensitivity relative to isolated utterances* [Poster presentation]. 3rd Interdisciplinary Conference on Voice Communication, Jena, TH, Germany.

Jun 29–Jul 2, 2026 **Xu, T.**, & Dellwo, V. *Perceived voice similarity within and across languages: A study of Cantonese–English bilingual speakers* [Oral presentation]. 13th International Conference on Language Variation in Europe, Lausanne, VD, Switzerland.

May 26–29, 2026 **Xu, T.**, Jiang, X., & Dellwo, V. *Dominant role of temporal–prosodic cues in sentence-level voice discrimination: Insights from interpretable machine learning* [Poster presentation]. Speech Prosody 2026, Philadelphia, PA, United States.

Dec 18–19, 2025 **Xu, T.**, Li, J., & Jiang, X. *Is argument structure processing syntactic or semantic in nature? Evidence from computational modeling on naturalistic fMRI* [Poster presentation]. Computational Psycholinguistics Meeting 2025, Utrecht, The Netherlands. https://openreview.net/forum?id=Yl7cxV5P4r

Jun 19–20, 2025 **Xu, T.**, Li, J., & Jiang, X. *Semantic processing of argument structure during naturalistic story listening: Evidence from computational modeling on fMRI* [Poster presentation]. 19th Lancaster Linguistics and English Language Postgraduate Conference, Lancaster, United Kingdom.

Apr 18, 2025 **Xu, T.**, Li, J., & Jiang, X. *Semantic processing of argument structure during naturalistic story listening: Evidence from computational modeling on fMRI* [Oral presentation]. 6th Annual Emory Undergraduate Linguistics Conference.

Invited Talks

Sep 8, 2026 *The role of speaker variability in statistical learning of word boundaries: An EEG study*. Colloquium Neuroscience of Speech & Language, University Hospital of Psychiatry Zurich, Zurich, ZH, Switzerland.

Sep 24, 2025 *Testing the multidimensional nature of voice perception: Past findings and future directions*. Colloquium Phonetics and Speech Sciences, University of Zurich, Zurich, ZH, Switzerland.

Teaching

Feb–May 2026 **Phonetics and Speech Sciences Reading Group** (Instructor: Volker Dellwo)
University of Zurich
- Coordinator of the Bayesian Methods subgroup

Apr 2026 **Experiments with speech** (Instructor: Volker Dellwo)
University of Zurich
- One guest lecture on speech synthesis

Sep 2023–Jan 2024 **Syntax** (Instructor: Jialei Zhu)
Shanghai International Studies University
- Teaching assistant

Sep 2023–Jan 2024 **Logic** (Instructor: Jialei Zhu)
Shanghai International Studies University
- Teaching assistant

Student Supervision

Jan 2026–present **Meng Yao**. *EEG evidence for the role of speaker variability in statistical learning of word boundaries*.
MA in Linguistics, University of Zurich
- Cosupervisor (Primary supervisor: Volker Dellwo)

Memberships & Service

2026–present **Member**
International Phonetic Association; International Speech Communication Association; Society for the Neurobiology of Language; Voice Communication Sciences Association

2026 **Reviewer**
Computational Psycholinguistics Meeting 2026

Jul 2026 **Session Chair**
13th International Conference on Language Variation in Europe, Lausanne, VD, Switzerland

Oct 2024–Jun 2025 **Member & Linguistics Olympiad Assessor**
Cambridge University Linguistics Society

Skills & Languages

Research & Data Analysis

- Python, R, MATLAB, and SPSS Statistics for statistical analysis and data visualisation
- Praat, VoiceSauce, and Adobe Audition for speech recording, processing, and acoustic analysis
- Natural language processing (NLP) and computational linguistics techniques
- Surveys, interviews, and behavioural, eye-tracking, EEG, MEG, and fMRI experiments

Languages

- Chinese: Mandarin (native), Wu (native), Cantonese (native)
- English (IELTS Academic Band 8), Japanese (basic), Korean (basic)